Candidate's Name……..…….

Signature……. stream…….

(Success only comes to those who prepare for it, therefore do your very best)

545/1
CHEMISTRY
Paper 1
April 2026
2 hours

For scorers' Use Only

Total Weighted Score		
Initials		

ANAKA INTERGRATED SECONDARY SCHOOL
Uganda Certificate of Education
END OF TERM I ASSESSMENT S.3
Paper 1
2 Hours

INSTRUCTION TO CANDIDATES:
This examination paper consists of **two** *Sections;* **A** *and* **B**.

It has **six** *examination items.*

Section **A** *has* **two compulsory** *items. Responses to section* **A must** *be written in the spaces provided.*

Section **B** *has two* **parts; I** *and* **II**. *Respond to only* **one** *item from* **each** *part.*

Responses to section **B must** *be written in the answer booklet(s)provided*

Respond to **four** *items in all.*

Responses to each item in section B should start on a fresh page.

Any additional ***item(s)*** *answered will* **not** *be scored.*

For scorers' Use Only

Item No.	Basic code	Weighted Score	Scorers' Initials
1	T		
	D		
	E		
2	C		
	X		
	I		
_____(3/4)	P		
	S		
	B		
_____(5/6)	N		
	M		
	Be		
Total Weighted Score			

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SECTION A

Respond to **all** *the items in this section in the spaces provided.*

Item 1

Among the requirements for a boarding section in schools are detergents which are mainly used for washing clothes. It has been discovered that most students use detergents **X** which waste a lot of water due to much lather formed which requires plenty of water for rinsing. This is because of the nature of the water used in the school. This has resulted into high rise in water bills.

The school authorities have advised the students to use alternative detergent **Y** for use since it does not waste a lot of water. However, they do not know its effectiveness and some of the side effect associated with the use of the product, you have been requested by your chemistry teacher to help them understand.

Figure 1 proposed detergent Y for use

Figure 2 detergents X used by most students

Task: help the school authorities to;

a) Know the categories of the proposed detergent and the most commonly used detergents by the students.

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b) Understand how the proposed detergents work in cleaning students' clothes.

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For scorer's use Only

Basic Code	T
Score	

c) Understand the dangers associated with long term use of the products.

_____

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For scorer's use Only

Basic Code	**D**
Score	

d) Evaluate the effectiveness of the cleaning products the students are using to the alternative detergents which was proposed.

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For scorer's use Only

Basic Code	**E**
Score	

Item 2

Compound **W** in feedstock is important in boosting milk production as well as meat product preservation.

W is obtained from salty waters of lake Katwe and experiments show that it contains about 39.3% sodium and the rest being chlorine.

Your class started the project of making feedstock for cattle on the school farm. Your classmates do not know the nature of the compound in the feedstock, its simplest formula, properties and consequences on the environment.

Task

Help the classmate and;

a) Explain the nature of compound W in the feedstock.

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For scorer's use Only

Basic Code	**C**
Score	

b) Determine the simplest formula of compound **W** in the feedstock hence name the compound. (Na=23, Cl=35.5)

c) Explain the properties of compound **W**, that makes it useful in its applications.

For scorer's use Only

Basic Code	**X**
Score	

d) Point out the environmental impact of compound W and how it can be mitigated.

For scorer's use Only

Basic Code	**I**
Score	

SECTION B
Part I

Respond to only **one** *item from this part in the answer booklet(s) provided.*

Item 3

Jane, a young entrepreneur, plans to set up a soap less detergent manufacturing plant near Akago river due to the availability of water for industrial use such as cooling machines.

She has assured the local leaders that the factory will be economically beneficial and operate in environmentally friendly mode.
However, the residents are concerned about how the process of production will be carried out and how the plant will affect their water sources and the surrounding environment.

Figure 3 soapless detergent factory producing fumes

The local council has invited you, as a chemistry student through the head of science club in your school to a brief sensitization workshop.

Task

Prepare a presentation to explain to the community.

Item 4

In Lungulu village, residents are struggling with fuel shortage and pollution caused by use of traditional energy sources such as firewood and charcoal, yet they rear cattle and poultry with plenty of waste they collect from the animals and use for only fertilizers. The district development officer has come with an idea of setting biogas production facility to provide a more reliable and cleaner source of energy for cooking and lighting.

While the locals don't know how the production plant will benefit them not just providing energy, the process of production and the possible impact of the plant to the environment and how it will be mitigated.

You have been invited to guide the community to address their concern by the district development officer in a sensitization workshop through the patron science club of your school.

Task

Make a write up of the message you will deliver in the sensitization workshop.

Part II

Respond to only **one** *item from this part in the answer booklet(s) provided.*

Item 5

Petroleum is a naturally occurring flammable liquid found beneath the earth's surface. There has been a gradual increase in the atmospheric temperatures in Uganda of recent due to excessive use of petroleum products as source of energy in industries, vehicles and most homes especially in rural areas. And this has left the locals worried.

The LC1 chairperson of one of the areas in Nwoya district has organized a meeting to sensitize the locals about these concerns.
You have been selected by the chairperson as a person who has knowledge of environmental conservation to help sensitize the locals about the issue at hand.

Task

Make a presentation that you will use during the meeting.

Item 6

In keyo, lamogi sub-county, Amuru district, most people survive on rock quarrying to earn a living. The tractor that ploughs gardens also releases a lot of fumes from exhaust pipes to the atmosphere. All these have caused keyo community members worried due to drastic weather change which is affecting Agriculture.

A head teacher of one of the nearby schools has tasked the school science club to organize a sensitization campaign aiming at mitigating the challenge. The theme of the campaign is "**my environment, my responsibility**". You have been chosen to speak to the rest of students regarding the theme. Figure **4** shows how rocks are being broken down into pieces and tractor producing fumes in figure **5**.

Figure 4 rock quarry using explosives and machines

Figure 5 tractors producing toxic fumes

END